Supplemental Material

Observable-Dependent Location of Finite-Time Amplification Maxima in Projected Quantum Dynamics

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This Supplement documents the numerical procedure behind Sections 6 and 7 of the main text. It covers the integration scheme, the status of each readout as a linear or nonlinear functional of the state, the linear observable family used to test Theorem 1 directly, the period-aware rate estimator, the acceptance criteria and the reasons two inherited criteria were not used, the convergence controls, and the contents of the code and data archive. Nothing here is required to follow the argument of the main text; it is provided so that every number in Sections 6 and 7 can be regenerated independently.

S1. Model, integration and observables

S1.1. Equations and parameters

The system is a parametrically driven anharmonic oscillator in the effective-hbar form

$$i\,\hbar_{\mathrm{eff}}\,\partial_t\,\psi = [\,-(\hbar_{\mathrm{eff}}^2/2)\,\partial_{xx} + V(x,t)\,]\,\psi\,, \qquad V(x,t) = \tfrac{1}{2}\,\Omega^2(t)\,x^2 + (\beta/4)\,x^4\,, \qquad \Omega^2(t) = \Omega_0^2\,[1 + A\cos(\omega_p\,t)]\,,$$

with $\Omega_0 = 1$ and $\beta = 1/6$ throughout. The control parameter is the pump detuning $\delta = \omega_p - 2\Omega_0$. The reference parameter point is $A = 0.20$, $\hbar_{\mathrm{eff}} = 0.25$; the four additional points of Section 7 are $(A, \hbar_{\mathrm{eff}}) = (0.14, 0.25)$, $(0.26, 0.25)$, $(0.20, 0.15)$ and $(0.20, 0.40)$.

Integration uses a symmetric split-operator propagator on a uniform periodic position grid over $[-L, L)$ with $L = 24$ and $M = 2048$ points, time step $dt = 3 \times 10^{-3}$, total record $T_{\mathrm{win}} = 30$, and moments recorded every eighth step. The initial state is the ground state of the static potential at $\Omega^2 = \Omega_0^2$, obtained by imaginary-time propagation, and is prepared independently of any observable.

S1.2. Observation maps, and which of them are linear

Three scalar signals are read from the same stored trajectory at each recorded time: the second moments $\langle x^2\rangle$ and $\langle p^2\rangle$, and $\sigma_{\max}$, the larger eigenvalue of the covariance matrix $C = [[\mathrm{Var}_x, \mathrm{Cov}_{xp}], [\mathrm{Cov}_{xp}, \mathrm{Var}_p]]$. Momentum moments are computed with the canonical momentum $p = -i\,\hbar_{\mathrm{eff}}\,\partial_x$, so that they carry explicit factors of $\hbar_{\mathrm{eff}}$; an inconsistent normalisation at this step changes inferred quantities by orders of magnitude and is one of the diagnostic-sensitivity examples retained in S5.

Theorem 1 of the main text is stated for linear expectation-value observation maps, $M_A(\rho) = \mathrm{Tr}(\rho\,A)$. The variances of a state are not of this form in general, so the status of each readout has to be established rather than assumed.

For the present system the two quadrature readouts are linear, and for a structural reason. The potential is even at every instant and the initial state is the ground state, so parity is conserved and the first moments vanish identically. Numerically, over the whole detuning scan,

$$\max|\langle x\rangle| = 5.3 \times 10^{-14}\,, \qquad \max|\langle p\rangle| = 6.4 \times 10^{-14}\,,$$

against second moments whose smallest values are $\langle x^2\rangle = 0.0426$ and $\langle p^2\rangle = 0.0512$ — thirteen orders of magnitude. The subtracted terms in $\mathrm{Var}_x = \langle x^2\rangle - \langle x\rangle^2$ are therefore at the level of the arithmetic itself, and

$$\mathrm{Var}_x = \langle x^2\rangle = \mathrm{Tr}(\rho\,x^2)\,, \qquad \mathrm{Var}_p = \langle p^2\rangle = \mathrm{Tr}(\rho\,p^2)$$

are equal within numerical precision. Both are admissible linear observation maps in the sense of Section 2 of the main text, and both fall directly under the hypotheses of Theorem 1. They are written as $\langle x^2\rangle$ and $\langle p^2\rangle$ throughout to keep this visible. The verification is performed by linear_maps.py, which stores the first moments alongside the second.

The third readout, $\sigma_{\max}$, is the larger eigenvalue of a matrix assembled from second moments and is therefore a nonlinear functional of the state. It is not covered by Theorem 1. It is retained because it is the quantity an experimentalist would most naturally extract, and it tests whether the mechanism the theorem isolates

for linear maps survives beyond the linear hypothesis. Every statement about σ_{max} in the main text is labelled as an empirical extension rather than an instance of the theorem.

S1.3. The linear observable family

Because $\langle x^2 \rangle$ and $\langle p^2 \rangle$ are linear expectation values, so is every member of the one-parameter family

$$A_\alpha = (1 - \alpha)\, x^2 + \alpha\, p^2 , \quad \alpha \in [0, 1] ,$$

which is the observable family of Section 2.5 of the main text with $A_0 = x^2$ and $B = p^2 - x^2$. Each member is Hermitian and positive semidefinite, so each is an admissible observation map and the corresponding signal is positive, as the log-linear estimator requires.

No further integration is needed to scan this family. Since $\langle A_\alpha \rangle = (1 - \alpha)\, \langle x^2 \rangle + \alpha\, \langle p^2 \rangle$, the entire family follows from the second moments already stored by linear_maps.py; linear_family.py evaluates it. This is why the direct test of Theorem 1 cost one additional pass over the detuning grid rather than a new scan for every observable.

The resulting locations of the maximum are monotone in α, from +0.0692 at α = 0 to +0.1073 at α = 1, a span of 0.0380. These endpoint values differ by less than 3.4×10^{-4} from the fixed-readout estimates of $\langle x^2 \rangle$ and $\langle p^2 \rangle$ reported in the main text, which is within the uncertainty of the quadratic peak-extraction procedure: linear_family.py fits the parabola over a fixed half-width of 0.03 on the 0.004 detuning grid, whereas robustness.py uses a half-width of 3.5 grid steps on the 0.008 grid. Applying the acceptance criterion to the two extreme members and to the midpoint $\alpha = 1/2$, 72 of the 96 estimator variants are accepted, and in every one of them the ordering is preserved; within this Aα-family envelope the smallest separation between extrema obtained from different members of the family is 0.0092. The corresponding figure for the three fixed readouts, where the nonlinear readout σmax is included, is 0.0062; the two are different quantities and are not directly comparable.

S2. The period-aware rate estimator

S2.1. Why an integer-sample boxcar is not sufficient

The quadrature variances oscillate strongly within each drive period. The natural smoother is a running average over one drive period $T_p = 2\pi/\omega_p$, but a running average over an integer number of grid samples does not coincide with one period, because T_p varies continuously with detuning while the sampling grid does not. At $\delta = 0$ the mismatch is about 0.7 per cent of a period. The uncancelled harmonic leaks into the log-linear fit and depresses the goodness of fit for Var_x and Var_p to $R^2 \approx 0.946$ while σ_{max}, which is an envelope, remains above 0.998.

S2.2. Exact one-period averaging

We remove the known frequency exactly rather than modelling it. The recorded signal is represented by a natural cubic spline, its analytic antiderivative I(t) is formed, and the running average

$$\bar{s}(t + T_p/2) = [\, I(t + T_p) - I(t) \,] \,/\, T_p$$

is evaluated in closed form at every sample. The averaged value is assigned to the centre of its interval; a constant time offset cannot change a log-linear slope, so this choice affects only which samples fall inside the fit window. No fitted parameter is introduced at any stage, so the extracted maxima cannot be attributed to model-dependent fitting. This is the reason an oscillatory fit component was not used instead.

S2.3. Why one pass is not enough, and why two are

For a growing signal the exact one-period integral does not annihilate the drive harmonics. Writing $s(t) = \exp(2\Lambda t)\, \Sigma_n\, c_n \exp(i\, n\, \omega_p\, t)$, the exact integral over one period gives

$$\bar{s}(t) = [(\exp(2\Lambda\, T_p) - 1)/T_p] \exp(2\Lambda t)\, \Sigma_n\, c_n \exp(i\, n\, \omega_p\, t) \,/\, (2\Lambda + i\, n\, \omega_p) ,$$

so harmonic n is suppressed relative to the secular term only in the ratio $|2\Lambda/(2\Lambda + i\, n\, \omega_p)| \approx 2\Lambda/(n\, \omega_p)$. With $\Lambda \approx 0.045$ and $\omega_p \approx 2$ this leaves about 4.5 per cent, which is precisely the residual observed: after one pass

the goodness of fit for Var_x and Var_p rises only to 0.965, and a spectral analysis of the fit residual places its dominant component at the drive frequency.

Each further pass applies the same factor again, so two passes leave a residual of order $(2\Lambda/\omega_\mathrm{p})^2 \approx 2 \times 10^{-3}$. The prefactor is a constant and the time origin shifts by $T_\mathrm{p}/2$ per pass; neither can affect a log-linear slope. Two passes is the minimum number that clears $R^2 \geq 0.98$ and is used as the reference estimator; the number of passes is one of the axes of the envelope of S4.

The effect of successive passes is shown below (common window [3.0, 13.0], reference parameter point, worst value over the whole detuning scan).

Passes	R^2 σ_max	R^2 Var_x	R^2 Var_p
1	0.9976	0.9651	0.9655
2	0.9978	0.9885	0.9891
3	0.9979	0.9939	0.9945

S2.4. Averaging in log space: tested and rejected

Since ln s(t) = $2\Lambda t + g$(t) with g periodic, an exact one-period average taken in log space would annihilate every drive harmonic in a single pass. This was implemented and rejected. Var_x passes close to zero at each period, and the logarithm amplifies those minima enormously; the goodness of fit falls to 0.925 for Var_x and 0.931 for Var_p, worse than the boxcar it was meant to replace. The additive structure that makes log space attractive in principle is defeated by the near-zeros of the quadrature variances in practice.

S3. Acceptance criteria

S3.1. The criterion used

A variant of the extraction procedure is accepted if the log-linear fit satisfies $R^2 \geq 0.98$ for all three observables at every detuning of the scan. This is a statement about whether a single exponential describes the fit window. It refers to no particular observable and to no property of the record as a whole, and is therefore comparable across observation maps and across parameter points.

S3.2. The saturation criterion and why it was not transferred

The companion calculation on the same system uses a window that ends before the signal reaches a fixed fraction (0.6) of its maximum over the recorded trajectory. That ratio measures proximity to saturation only if the record actually flattens. It does not here. At the reference parameter point the period-averaged log-slope has fallen only to 45 per cent of its early value by the end of Twin = 30, so the ratio in fact measures the fraction of the recorded growth and therefore depends on Twin.

Because the growth rate scales roughly as A/4, the same physical window is scored very differently at different pump amplitudes. At A = 0.14 with Twin = 30 the criterion returns 0.65 even for a window ending at t = 9 and rejects every window on the grid; extending the record to Twin = 52 does not repair this, since the trajectory has still not flattened by then. The criterion is not comparable across parameter sets and was therefore not used.

S3.3. The window-spread criterion and why it was not transferred

The same companion calculation also uses a half-window slope-disagreement criterion, $|s_\mathrm{first} - s_\mathrm{second}| \,/\, |\mathrm{mean}| \leq 0.10$, which is intrinsic to the window and does not depend on the record length. It was nevertheless not used, for a different reason: it was formulated for the covariance top eigenvalue, which is an envelope.

The quadrature variances carry a slow modulation arising from the rotation of the squeezing axis under detuning. This modulation is not at a harmonic of the drive, so no amount of period averaging removes it, and it makes the two half-window slopes disagree by about 40 per cent at every window length tested, while σ_max stays between 0.03 and 0.18. Imposing the 0.10 threshold accepts zero of the 96 estimator variants. The criterion does not test window quality in this setting; it tests whether the observable happens to be an envelope, and so privileges one of the three compared observation maps.

Neither discarded criterion can be repaired without electing one of the compared observables as the arbiter of the window, which is inadmissible in a comparison whose entire content is the difference between observation maps.

Window end t_1	spread $\sigma_{\max}$	spread Var_x	spread Var_p
7	0.030	0.393	0.337
9	0.056	0.405	0.394
11	0.098	0.425	0.420
13	0.153	0.431	0.413

Table S1. Half-window slope disagreement at the reference parameter point, two averaging passes, window start $t_0 = 3.0$, worst value over the detuning scan. The quadrature variances never approach the 0.10 threshold at any window length; $\sigma_{\max}$ does.

S4. Envelope and convergence controls

S4.1. Estimator envelope

The estimator was swept over 96 variants formed from four window starts (2.5, 3.0, 3.5, 4.0), six window ends (11 through 16) and four numbers of averaging passes (1 through 4). Seventy-two satisfy the acceptance criterion. Over those 72 variants the extracted locations vary as follows.

Quantity	mean	min	max
δ_x	+0.0763	+0.0565	+0.0896
δ_σ	+0.0970	+0.0888	+0.1033
δ_p	+0.1061	+0.1002	+0.1112
$\Delta_{\sigma x}$	+0.0207	+0.0132	+0.0326
$\Delta_{p\sigma}$	+0.0091	+0.0062	+0.0164
Δ_{px}	+0.0298	+0.0211	+0.0488

Table S2. Envelope of the extracted maxima and separations over the 72 accepted estimator variants at the reference parameter point. The ordering $\delta_x < \delta_\sigma < \delta_p$ holds in every accepted variant without exception; for these three fixed readouts the smallest separation across accepted estimator variants is 0.0062.

The absolute positions are not converged in the number of averaging passes: each additional pass moves them by roughly 0.003. This is not a numerical defect. The kernel width is npass × T_p, and smoothing over a finite window interacts with the time-curvature of ln s(t), which itself varies with detuning because the growth is faster and its deceleration stronger near resonance. It is an intrinsic finite-window effect, and it acts identically on the three observables, so it cannot generate a splitting between them. Absolute locations should accordingly be reported together with the procedure that produced them; the separations are the procedure-independent content.

S4.2. Grid, time-step and interpolation controls

Grid and time step. At $\delta = 0$ the three extracted rates agree to 4×10^{-6} between (M, dt) = (4096, 1.5×10^{-3}), (2048, 1.5×10^{-3}) and (2048, 3×10^{-3}). Repeating the fine scan with the time step halved moves the extracted maxima by at most 4×10^{-4}, two orders of magnitude below the separations.

Detuning grid. Decimating the detuning grid by a factor of two and re-extracting, over 400 random offsets, changes the maxima by 0.00000 ($\sigma_{\max}$), 0.00011 (Var_x) and 0.00034 (Var_p).

Interpolation. Varying the half-width of the quadratic fit used for the sub-grid argmax from three points to ±0.06 in δ moves the maxima by at most 0.006, and by at most 0.0003 for $\sigma_{\max}$.

S5. Extraction-rule and normalisation sensitivity

The following two observations are not instances of the theorem, which concerns the observation map, but they arise in practice at comparable magnitude and are recorded here because the practical protocol of Section 8 is designed to catch them.

A fit window placed inside the interval in which the tangent direction has not yet stabilised returns a rate depressed by roughly fifteen per cent relative to the converged value, with no internal indication that anything is wrong. An inconsistent normalisation of the momentum moments between two branches of a comparison — wavenumber moments on one side, canonical-momentum moments on the other — alters an inferred residual by nearly two orders of magnitude. In each case the reported number is reproducible and internally consistent within its own procedure. Both examples are taken from Sec. III of the AMT companion study of this system, Ref. [23] of the main text (Zenodo, DOI: 10.5281/zenodo.21859415), where they arose in the validation of the same grid integrator.

These are levels of the diagnostic pipeline at which no general analytical invariance criterion is presently known, and for which robustness must be established numerically.

S6. Code and data

All scripts, cached results and raw signals are archived and available; the archived version corresponding to the calculations reported here carries a persistent identifier given in the Data Availability statement of the main text. The contents are as follows.

File	Contents
`SM_grid_engine.py`	Split-operator grid engine: grid, imaginary-time ground state, evolution with $\hbar_{\mathrm{eff}}$, covariance moments in canonical variables
`delta_scan.py`	Detuning scan: one integration per δ, three observables from the same trajectory, common estimator and window, sub-grid argmax
linear_maps.py	Verification that the first moments vanish, and storage of the three independent second moments $\langle x^2\rangle$, $\langle p^2\rangle$ and $\langle(\mathrm{xp+px})/2\rangle$, all linear in ρ
linear_family.py	Location of the maximum along the linear family A_α, evaluated from the stored moments without re-integration
`estimator.py`	Period-aware estimator: cubic spline, analytic antiderivative, exact one-period running average, multi-pass variant
`runner.py`, `store_signals.py`	Staged execution of the coarse and fine scans; storage of raw signals
`envelope.py`	Sweep over the 96 estimator variants with the acceptance criterion applied
`compare_estimators.py`	The three acceptance checks performed when the estimator was changed
`robustness.py`	Repetition of the scan at the five points in (A, $\hbar_{\mathrm{eff}}$)
`dtcheck.py`	Time-step convergence control
`mkfig.py`, `mkfig_robustness.py`	Generation of Figures 1 and 2 of the main text
`signals.npz`	Raw three-readout signals on the fine detuning grid; regenerates every table and every figure without re-integration
moments.npz	First and second moments on the same grid; the source for the linear observable family
`coarse.json`, `fine.json`, `robustness.json`	Cached scan results

Python 3.11 with numpy ≥ 1.24, scipy ≥ 1.10 and matplotlib ≥ 3.7. Execution order: `runner.py` (coarse), `runner.py` (fine), `store_signals.py`, `envelope.py`, `dtcheck.py`, `robustness.py`, then the two figure scripts.